

Tyler Stowell
tylerrstowell@gmail.com | [LinkedIn](#) | [Personal Website](#)

EDUCATION

Brigham Young University – Bachelor of Science, Electrical Engineering April 2027

- Minor: Physics
- Cumulative GPA: 3.99 / 4.0

EXPERIENCE

CamachoLab - Research Assistant Sep 2021 - Jun 2022, Jun 2024 - Present

- Designed, fabricated, packaged, and tested integrated silicon photonics devices in collaboration with employees at Nokia
- Deployed a quantum random number generator on site at Oak Ridge National Laboratory for a quantum key distribution system
- Wirebonded photonic chips in BYU's cleanroom and edge coupled them for characterization
- Designed couplers and crossover waveguides for an active green machine PIC with collaborators at the University of Arizona, UMD, and Tyndall National Institute

Center for Quantum Networks - Quantum Engineer (Contract) May 2026 - Aug 2026

- Built a quantum frequency conversion system from scratch for silicon vacancy quantum memory experiments with ~75% internal quantum efficiency
- Developed a cavity-coupled color center simulator that reproduces published experimental results to prepare for experimental work on a system ordered from IonQ (manuscript in preparation)
- Restored a pulsed laser system (water cooling, N2 purge, alignment) and supported alignment of the silicon vacancy quantum memory setup

Harsch Systems - Design Engineer June 2020 - Aug 2021

- Designed two generations of an RP2040-based PCB for industrial control of 4-20 mA and 0-10 V inputs and outputs
- Wrote a Linux kernel driver so Raspberry Pis could meet strict SPI timing requirements for industrial I/O applications under heavy load
- Brought up embedded Rust on the then-unsupported RP2040, writing register level GPIO, ADC, and PWM test programs

SERVICE & LEADERSHIP

The Church of Jesus Christ of Latter-day Saints – Volunteer July 2022 – June 2024

- Served a two-year, Spanish-speaking volunteer mission in southern Argentina, teaching and doing community service
- Led and trained groups of volunteers as district and zone leader
- Served as the primary financial and housing secretary for 6 months, managing all finances and housing contracts for ~150 volunteers

PUBLICATIONS

Integrated Differential Conjugate Homodyne Detection for Quantum Random Number Generation – arXiv Preprint (Under review)

- Wrote FPGA code for Toeplitz hash-based entropy extraction from quantum shot-noise limited measurements at a rate of 12 Mbps

Quantum Conference Key Agreement Using a Photonic Integrated Circuit Green Machine – Conference Article (Optica Quantum 2.0)

- Characterized a PIC capable of superadditive measurements of quantum signals
- Built a digital twin and QKD simulations (decoy states, finite-key effects) to compute conference key rates

Device-Aware Quantum Photonic Simulator for Gaussian States – Conference Paper (Frontiers in Optics + Laser Science 2022)

- Contributed quantum simulation features to the Simphony photonic circuit simulator, later extending it to sums of Gaussian states (e.g. cat states)

SELECTED PROJECTS

Single-Wheel Robot (2025)

- Designed, 3D-printed, wired, and programmed a modular robot from scratch that converts between two-wheel and single-wheel balanced modes

Laser Tag System (2026)

- Designed an optical transceiver board (LED transmitter, photodiode receiver, filtered amplifier) that placed in the top 10% of the class for signal-to-noise ratio

Real-Time Noise Floor Computer (2021)

- Wrote code that estimated the noise floor of signals from the adaptive-optics wavefront sensor on the Daniel K. Inouye Solar Telescope in real time

SKILLS

Cleanroom Experience

- Fabricated MOSFETs, photodiodes, and a diode laser in BYU's cleanroom

Optics, Photonics, and PIC Design

- Lumerical FDTD, GDSFactory, Free Space Optics Design, PIC Testing, Edge Coupling, Wirebonding

Programming and Hardware

- Python, C, Rust, FPGA (Xilinx Vivado), KiCAD PCB Design, Linux Kernel Drivers

Engineering Teams

- Hobby Robotics, BYU Mars Rover Team, BYU Rocket Club

Languages

- English (native), Spanish (fluent)